

IT 230 Coding Activity Submission Template

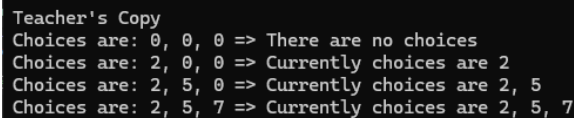
Submit your work on the coding activities for Modules One, Two, Three, Four, and Six in this document. In addition to this document, you should submit a ZIP file containing all your Visual Studio project files and source code that can be run in Visual Studio on a different computer.

For each coding activity, complete the following steps:

- Download and rename this document to meet the file naming conventions requested in the assignment instructions.
- Fill in the required information below by replacing the bracketed text with the relevant information.
- Submit this document and your ZIP file for grading and feedback. Your ZIP file should follow the same naming conventions.

Document your work in the coding activity by completing each of the following items:

1. Provide a screenshot of the output that resulted from running your program successfully in Visual Studio. See the coding assignment instructions for an example of what should be included in the screenshot. Your screenshot must include the following elements:
 - a. Your last name as the first printed text on the screen
 - b. Verification that the program is fully functioning and data results are accurate for the given problem



```
Teacher's Copy
Choices are: 0, 0, 0 => There are no choices
Choices are: 2, 0, 0 => Currently choices are 2
Choices are: 2, 5, 0 => Currently choices are 2, 5
Choices are: 2, 5, 7 => Currently choices are 2, 5, 7
```

[Insert your screenshot here.]

2. Copy and paste the source code text you wrote for this assignment from the *.cs file into the space below. Only providing the *.cs files or a screenshot does not meet the requirements for

this part of the assignment. Code should be logically organized. It should also follow proper syntax and conventions noted in the Coding Activity Guidelines and Rubric.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace DebugFixIFStmt
{
    class Program
    {
        static void Main(string[] args)
        {
            (new Program()).run();
        }

        void run()
        {
            int firstChoice = 0, secondChoice = 0, thirdChoice = 0;

            System.Console.WriteLine("Teacher's Copy");

            firstChoice = 0; secondChoice = 0; thirdChoice = 0;
            WriteCurrentChoices(firstChoice, secondChoice, thirdChoice);

            firstChoice = 2; secondChoice = 0; thirdChoice = 0;
            WriteCurrentChoices(firstChoice, secondChoice, thirdChoice);

            firstChoice = 2; secondChoice = 5; thirdChoice = 0;
            WriteCurrentChoices(firstChoice, secondChoice, thirdChoice);

            firstChoice = 2; secondChoice = 5; thirdChoice = 7;
            WriteCurrentChoices(firstChoice, secondChoice, thirdChoice);

            Console.ReadKey();
        }

        void WriteCurrentChoices(int firstChoice, int secondChoice, int thirdChoice)
        {
            if (firstChoice == 0)
                Console.WriteLine(
                    "Choices are: {0}, {1}, {2} => There are no choices",
                    firstChoice, secondChoice, thirdChoice);

            else if (secondChoice == 0)
                Console.WriteLine(
                    "Choices are: {0}, {1}, {2} => Currently choices are {0}",
                    firstChoice, secondChoice, thirdChoice);

            else if (thirdChoice == 0)
                Console.WriteLine(
                    "Choices are: {0}, {1}, {2} => Currently choices are {0}, {1}",
                    firstChoice, secondChoice, thirdChoice);
        }
    }
}
```

```
else
    Console.WriteLine(
        "Choices are: {0}, {1}, {2} => Currently choices are {0}, {1}, {2}",
        firstChoice, secondChoice, thirdChoice);
    }
}
```

3. Show that you understand the task by explaining the design of your program in the space below. Include the process and steps you took to write your code. Explain how you arrived at the solution to the problem and completed the activity.

To complete this activity, I opened the DebugFixIFStmt project in Visual Studio and attempted to compile and run the program. I reviewed the code and identified several errors in the conditional statements that prevented the program from producing the required output.

I corrected the comparison operators, fixed the conditional logic, and verified that the program displayed only the non-zero choices as required by the assignment. After making the corrections, I rebuilt and tested the application using all provided scenarios. The output matched the expected results, confirming that the program was functioning correctly.

4. Reflect on your learning experience and what you learned from completing the activity.

This activity improved my understanding of debugging and conditional statements in C#. I learned how small syntax mistakes, such as using the wrong comparison operator, can prevent a program from functioning correctly. I also gained experience reading error messages and tracing program logic to identify the source of problems.

Testing the program after each correction helped me understand the importance of verifying code changes and ensuring that all possible conditions produce the expected results. This activity strengthened my confidence in debugging and working with if, else if, and else statements.